

Figure Catalog

CytoCV Research Documentation

Purpose

This catalog identifies the current CytoCV diagram set for manuscript figures, supplementary figures, reports, appendices, and collaborator packets. It uses the same public terminology as the current application and documentation: DIC, Blue, Red, Green, .dv, stack TIFF, plugin-based statistics, Display/Dashboard review, CSV/XLSX export, and saved/transient runs.

Recommended Primary Figures

Figure	Source	Recommended use	Caption focus
Figure 1. System architecture	docs/diagrams/01-system-architecture.png	Overall software architecture	Browser workflow, Django services, job processing, scientific image-analysis services, and persistence boundaries.
Figure 2. End-to-end workflow	docs/diagrams/02-end-to-end-workflow.png	Methods overview	.dv or stack TIFF upload through validation, channel mapping, DIC-guided segmentation, plugin measurement, review, saved/transient storage, and CSV/XLSX export.
Figure 3. Cell analysis flow	docs/diagrams/03-cell-analysis-flow.png	Per-cell measurement section	Retained DIC masks, canonical fluorescence contours, puncta modes, contour intensities, CEN-dot classification, biorientation, nuclear-cell-pair intensity, and legacy Blue modules.
Figure 4. Data model	docs/diagrams/11-data-model.png	Reproducibility or implementation appendix	Relationships among uploaded runs, upload-preparation jobs, analysis jobs, previews, segmented outputs, and per-cell statistics.

Supplementary Figures

Topic group	Figure sources	Typical use
Artifacts and retention	04-artifact-lifecycle.png; 13-run-ownership-retention-state.png	Explain saved/transient result storage, generated artifacts, and cleanup behavior.
Channel and validation rules	05-plugin-channel-map.png; 06-upload-validation-flow.png; 07-scale-channel-resolution.png	Document required-channel logic, metadata interpretation, TIFF/DV scale handling, and pixel/micron conversion.
Processing and output	08-preprocess-inference-flow.png; 09-segmentation-output-flow.png; 10-display-export-flow.png	Provide detailed workflow explanation beyond the main narrative figure set.
Progress and cancellation	12-progress-cancellation-state.png	Explain worker-backed progress phases, cancellation, terminal states, and failure handling.
Access and legacy context	14-authentication-account-flow.png; 15-legacy-blue-measurements.png	Supplementary account-flow overview and backward-compatible legacy Blue measurement context.

Figure Usage Notes

- Use Figures 1-4 for the primary manuscript narrative unless the target journal requests fewer software architecture figures.
- Use supplementary figures for reviewer questions about validation rules, workflow control, generated artifacts, access control, or legacy behavior.
- Treat CEN-dot, biorientation, puncta, and nuclear-cell-pair outputs as software-generated measurements unless biology-side validation is described separately.
- Keep figure captions aligned with current logical channel terms rather than older fluorophore-specific or instrument-only terms.
- Regenerate all PNGs after any .mmd change and verify the images open in ordinary PNG viewers before submission.

Maintenance Note

Editable diagram sources and rendered PNG assets are maintained in docs/diagrams/. The render script in that folder validates PNG compatibility so corrupted raster files cannot silently remain in the repository.